Power System Stability and Control Analysis: Application on the Isolated PIAT Algerian Network

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***Abstract*—This paper deals with power system stability and control analysis using EUROSTAG 4.5 software. The main interest was to study the dynamic behavior of power system in view of major incidents and vital scenarios such as transient stability and frequency stability. In addition, the design of a robust control strategies like the Automatic Generation Control (AGC) and the Under Frequency Load Shedding (UFLS) was envisaged. An application on the IEEE 10 bus test system and on the islanded 57 bus Algerian grid called Pole InSalah-Adrar-Timimoun (PIAT) was performed. The dynamic modeling was carried out using the EUROSTAG Macro-Block tool. A series of events such as three-phase short circuit and load variation were presented during the simulation. The main attention in this work was the modeling and the simulation of alternating fast and slow phenomena such as the rotor oscillations and the centralized control for enhancing power system operation and control.**

Keywords—Frequency Stability, Transient Stability, Automatic Generation Control (AGC), Under Frequency Load Shedding (UFLS), Algerian Network, EUROSTAG.

I. INTRODUCTION

The current electrical power grids are characterized by interconnections between many areas or countries. This interconnection has created new problems of static and dynamic stability. The high sensitivity of interconnected power grids to grid disturbances, such as short circuit faults, load variation and frequency fluctuations, often leads to disconnection between interconnected grids. The current trend is for interconnection to be part of network tuning to ensure stability in frequency variations or an imbalance between production and consumption [1-2].

The issue of electric power system stability is an old and vital problem that have been reported during several years in a lot of published papers and reports of high quality on the subject by CIGRE and IEEE. According to the literature [3-6], power system stability can be defined as the ability of the system to remain in a steady state after disturbance. This latter is defined as a fast variation or a sudden change of the system parameters. Power system disturbances are classified in two types: the first one is the Large Disturbances **(**LD**)** such as the three-phase short circuits, disconnections of generators or loads, where the second one is the Small Disturbances **(**SD**)** such as small variations in the load or power supply [6-7].

A literature survey on the topic of power system transient stability revealed that some models have been proposed to analyze and improve system stability using recent developed devices and techniques [8-9]. However, the transient stability analysis may involve the calculation of Critical Clearing time (CCT) for a given fault.

While, the CCT is defined as the maximum allowable value of the clearing time for which the system remains to be stable. In reference [10] authors have proposed a model in MATLAB/Simulink to analyze the transient stability of a multimachine power system. In reference [11]. The authors have used a transient energy function for real-time prediction and control of transient stability, where, in reference [12], the authors have proposed the use of WAMS system for transient instability detection.

On the other hand, the demand for electricity is rising phenomenally especially which is leading to operation of the electrical grid at its limit. However, in case of any imbalance between supply and power demand, the network may lose his stability. In order to satisfy the electric power demand and ensure a good quality power frequency stability is envisaged as one of the vital keys in the grid [13-14]. The frequency stability can be defined as the ability of the grid to maintain the nominal frequency following an imbalance between supply and load. In view of this issue, many works have been published during the last years in the scope of frequency stability [15-17], where the use of optimization algorithms and artificial intelligence has been of main tasks.

In recent years, various commercial software packages such as DIgSILENT, ETAP, EMTP-RV and PSS/E have been widely used in the literature for power system simulation and analysis. In reference [18] the PSAT toolbox was used to design an intelligent automatic generation control (AGC) for optimal use in Smart Grids. In reference [19], authors have proposed a comparative study of Load Flow Simulation Using PSAT and Power World software. In reference [20], authors have proposed the use of ETAP software to analyze the transient stability of an interconnected network, where, in reference [21], authors have proposed the use of power world simulator for multimachine Transient stability analysis. In reference [22], power system stability analysis of an offshore wind farm connected to Taiwan electrical grid was presented using DIgSILENT software.

In view of the above discussion, the purpose of this paper was to analyze the transient and frequency stability of the isolated PIAT Algerian power system during disturbances. In addition, the implementation of the AGC and UFLS systems have been also considered to enhance system stability using a powerful tool for dynamic simulation named EUROSTAG.

The rest of the paper is organized as follow. Section II presents a brief description of the EUROSTAG software. Section III presents the simulation results of the transient stability analysis. Section IV is devoted to the presentation of the simulation results of the frequency stability, AGC and UFLS study. Finally section V concludes the paper.

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II. EUROSTAG SOFTWARE

In this section, the EUROSTAG software and its advantages in the field of electrical network analysis and dynamic simulation was presented. In addition, the dynamic modeling of voltage and speed control systems using the Macro-Block tool of EUROSTAG was also presented.

A. EUROSTAG PRESENTATION

EUROSTAG is a professional software for time domain simulation and efficient tool for dynamic analysis. It is developed by RTE (French Transmission System Operator) and TRACTEBEL, running under PC or workstation. EUROSTAG consists of a powerful tool, which allows it to define the level of modeling adapted to the type of study performed. Also, is based on a powerful algorithm that ensures the accuracy of results and the minimization of computation time, while a complete modeling is maintained in all phases of the calculation [23-24].

The EUROSTAG software also allows the study of very advanced topics such as: the black-out and voltage collapse, defense plans, FACTS, HVDC, wind energy. EUROSTAG is based on a variable time-step algorithm, which notably allows the study and simulation of stability and the analysis of the short and slow dynamics (frequency domain) of the power system as shown in Fig.1[25-26].

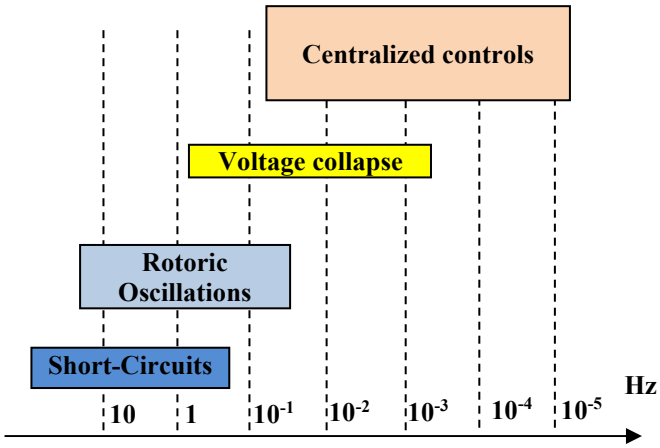


Fig. 1. The range of EUROSTAG applications [25].

The used models in this work are developed with EUROSTAG Macro-Block option as shown in Figs.2 to 5. Fig.2 presents a simplified Steam Governor-Turbine model. Fig.3 show the IEEE Gas Governor-Turbine. The voltage regulators AVR 3 and AVR 4 associated to the Algerian power generation station are presented in Figs.4 and 5 respectively.

B. Governor-Turbine Model

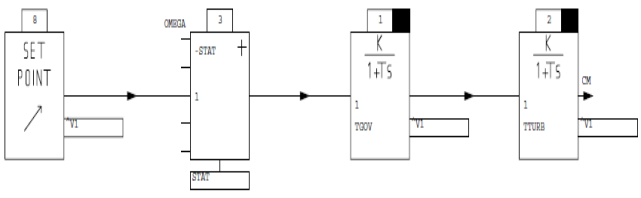


Fig. 2. Single Steam Governor-Turbine model under EUROSTAG.

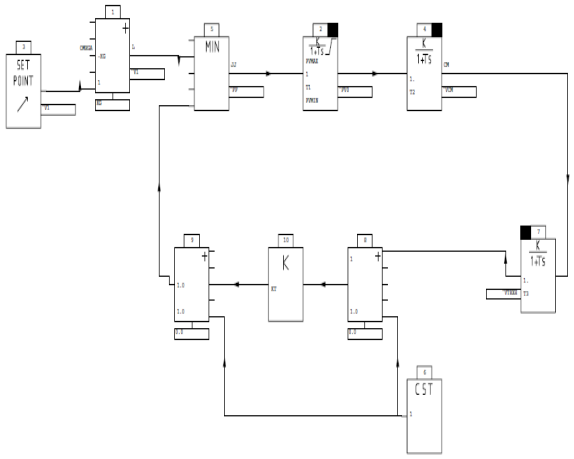


Fig. 3. IEEE Gas Governor-Turbine model under EUROSTAG.

C. AVR Model

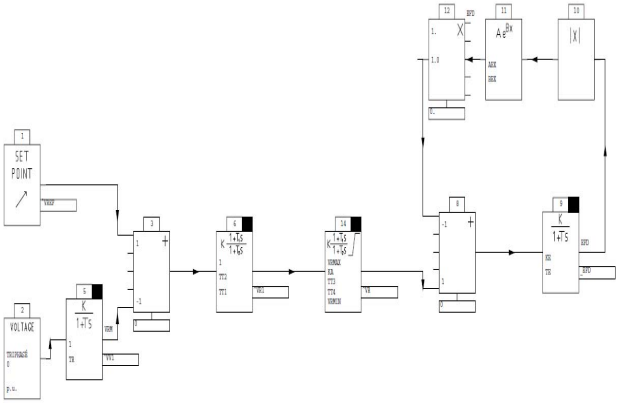


Fig. 4. Developed model of a voltage regulator AVR 3 under EUROSTAG.

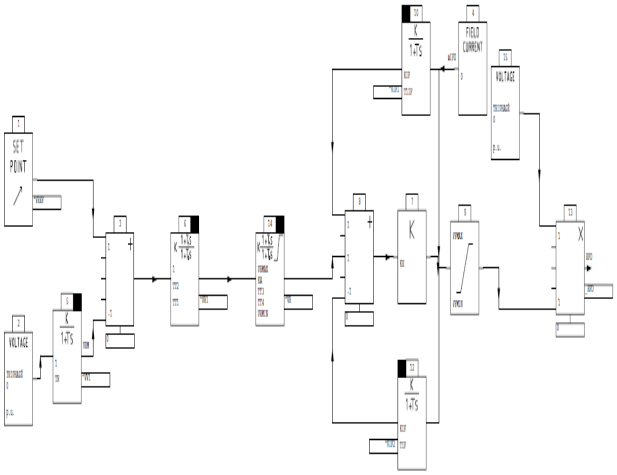


Fig. 5. Developed model of a voltage regulator AVR 4 under EUROSTAG.

III. TRANSIENT STABILITY SIMULATION RESULTS

In this section, the transient stability simulation results of the interconnected two areas IEEE 10 bus test system and the isolated Algerian network named Pole In Salah-Adrar-Timimoun (PIAT) are presented. A three-phase short circuit fault was considered as system disturbance, where the calculation of Critical Clearing time (CCT) was involved for each fault.

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A. IEEE 10 bus

In this part, the interconnected two areas IEEE 4 machine, 10 bus power system was used for the simulation as shown in Fig.6. This system was simulated for a stable and unstable scenarios. during the stable scenario a three-phase short circuit was applied during 200 ms at bus 11, and during 640 ms at the same bus in case of unstable scenario, where, the simulation was carried out during 30 s. The transient responses for the rotor angel and voltage in case of stable scenario are presented in Figs.7 and 8 respectively, while, in case of unstable scenario, the rotor angel and voltage responses are presented in Figs.9 to 10 respectively.

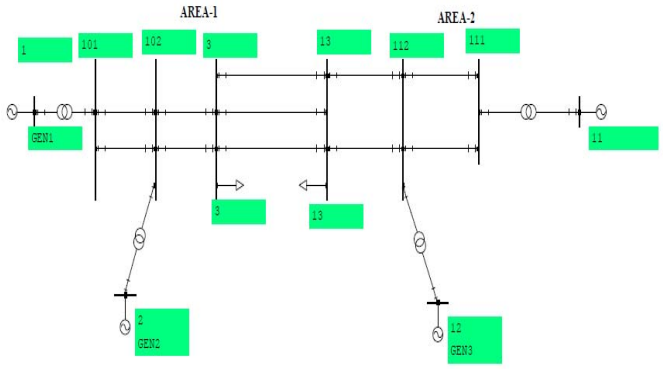


Fig. 6. IEEE 10 bus, 4 machine test system model in EUROSTAG.

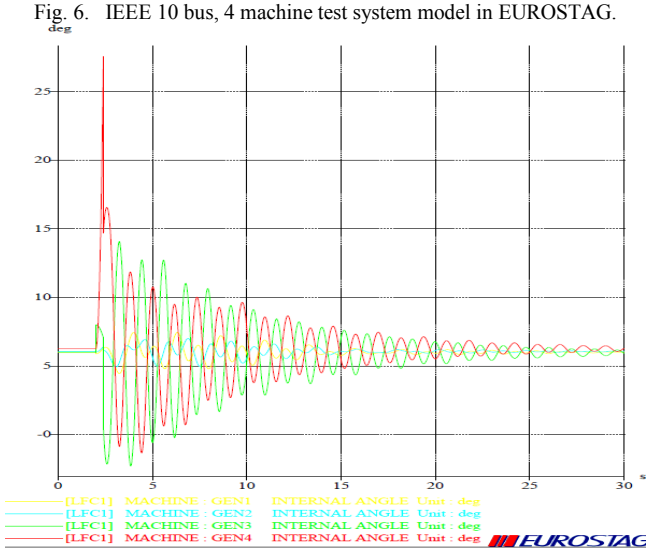


Fig. 7. Rotor Angel following a three-phase short circuit fault.

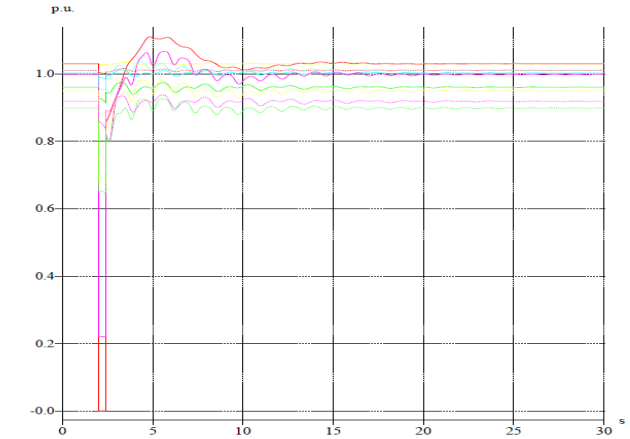


Fig. 8. Voltage following a three-phase short circuit fault.

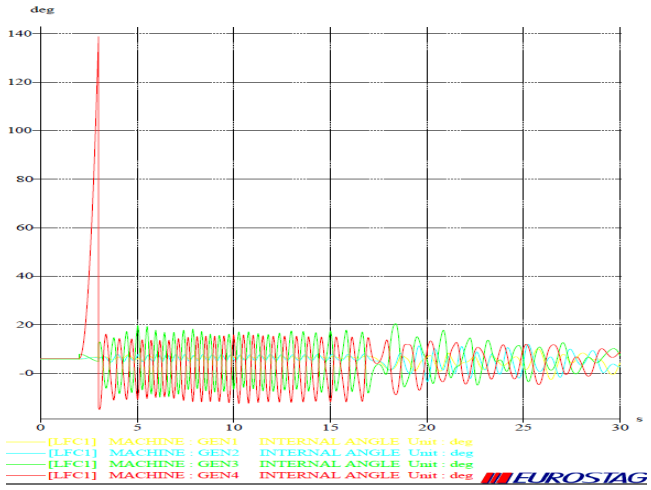


Fig. 9. Rotor Angel following a three-phase short circuit fault.

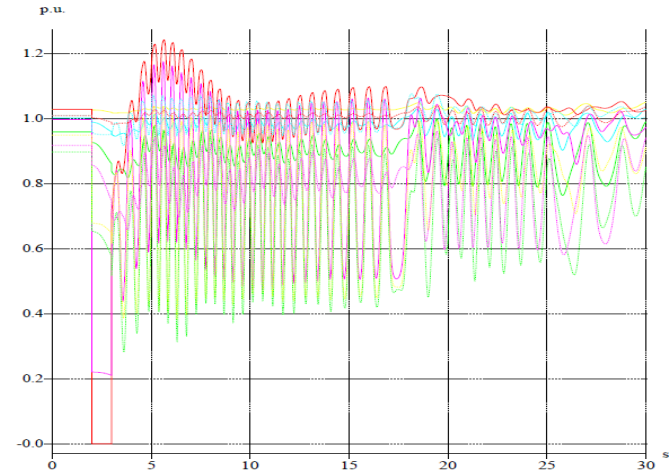


Fig. 10. Voltage following a three-phase short circuit fault.

B. PIAT Network

In this part, the interconnected three areas 25 generators, 57 bus isolated Pole In Salah-Adrar-Timimoun (PIAT) Algerian network was investigated. This system was devloped under EUROSTAG software as shown in Fig.11. A three-phase short circuit was applied during 100 ms at bus 10, and the simulation was carried out during 20 s. The transient responses for the rotor angel and frequency are presented in Figs.12 and 13 respectively.

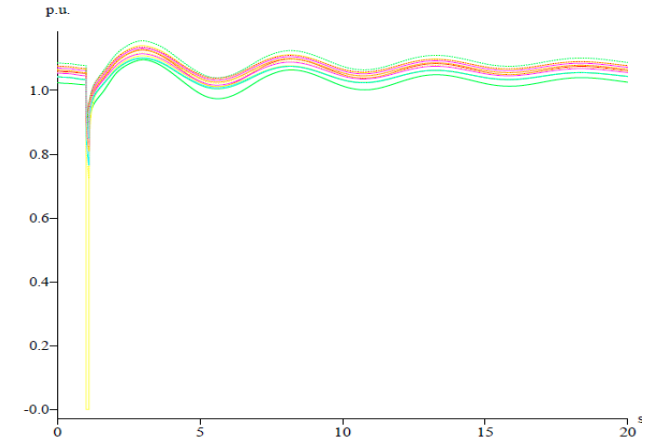


Fig. 12. Voltage following a three-phase short circuit fault.

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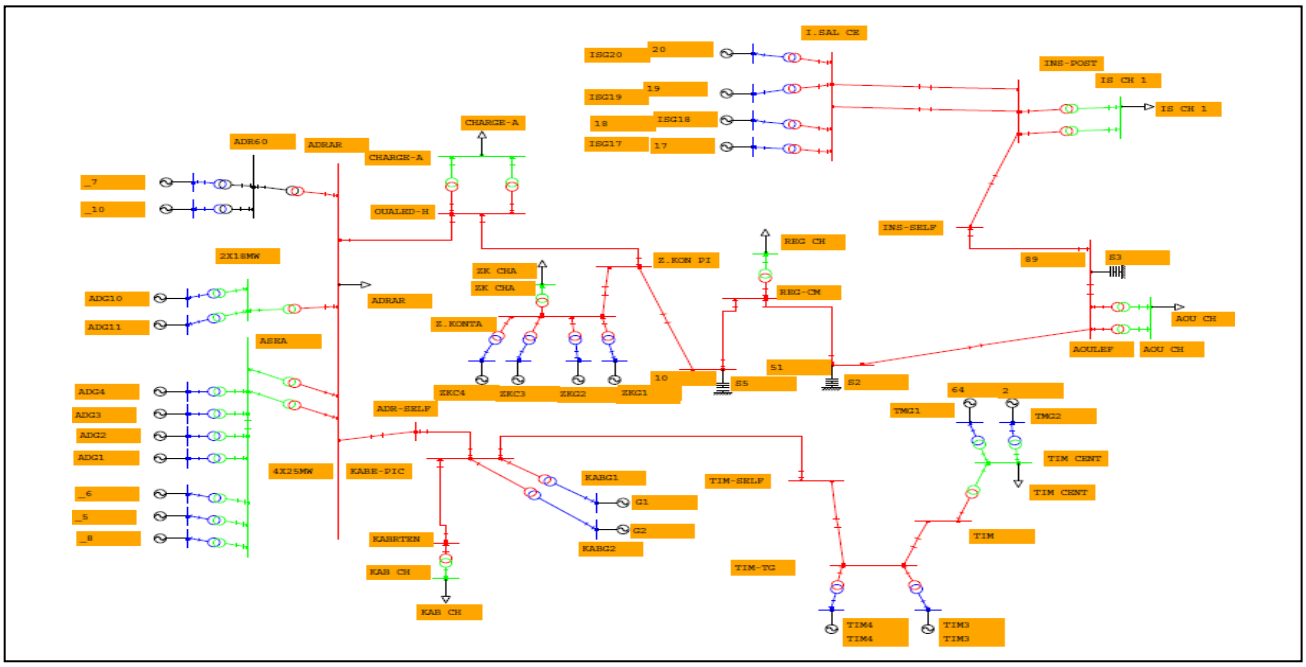


Fig. 11. Isolated PIAT Algerian Network.

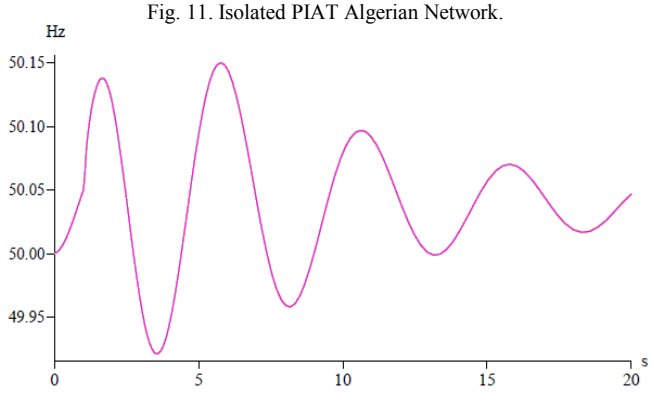


Fig. 13. Frequency following a three-phase short circuit fault.

C. Results Discussion

From the obtained results presented in the above figures it can be seen that for the variation of the internal angles of the machines presented in Fig.7 which are stable in the period before the disturbance (t <1s) and keep their initial values until the appearance of the defect at t=tf = 1s, where the internal angles increase, and after the defect elimination time (te=$1.2s$), the curves of the internal angles start to oscillate around of their initial values, this is the case of a stable system. In contrast, in Fig.9 the internal angles continue to increase after the fault has been eliminated and reach unacceptable values, the system in this case is unstable. Figs.8 and 12 show the curves of the bus voltages. It can be observed that the bus voltages are stable before the appearance of the fault, where the voltages decrease to very low values (tend towards 0), then the curves of the voltages resume their initial values that before the appearance of the fault with small oscillations around the nominal values, and the system can considered as stable. On the other hand, Fig.10 presents the strong variation of the bus voltages, where these curves are presented by large oscillations, and the system is considered as unstable. Finally, Fig. 13 show the frequency following the three-phase short circuit fault for a stable scenario.

IV. Frequency Stability and AGC/UFLS Results

In this section, power system frequency stability with AGC and UFLS control was envisaged and applied on the same test systems of the previous section. The simulation was performed under system load variation.

A. IEEE 10 bus

In this part, the frequency stability in the IEEE 10 bus network was analyzed following 10% and 50% load increase at bus 6 as shown in Figs.14, and 15 respectively. As shown in Fig.14 the frequency decrease from 50 Hz towards 49.86 Hz, after that start to oscillate around a new frequency value of 49.94 Hz using primary regulation. On the other side, in case of 50 % load increase, the frequency decrease at 48.6 Hz with multi-oscillations in the system frequency, which is undesirable case. This last case led to the use of the second regulation system AGC or the load shedding (UFLS). As a scenario, Fig.16 represent the partial load shedding applied at t=60s. It can be observed from Fig.16, that after the load shedding, the system frequency start to oscillate around a two novel frequency, the first one is 50.2 Hz for area-1 that includes generator-1 and generator-2, where, the second frequency is about 49 Hz in area-2 that includes generator-3 and generator-4.

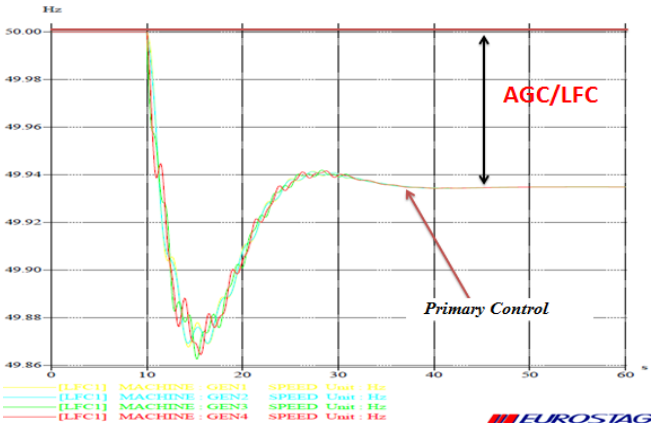


Fig. 14. Frequency following 10% load increase at bus 6.

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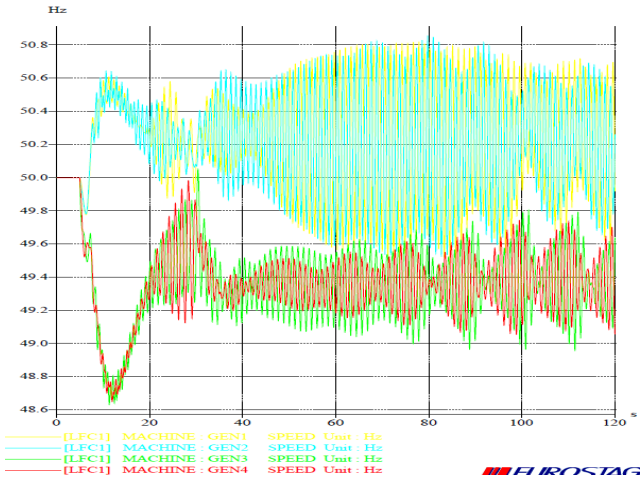


Fig. 15. Frequency following 50% load increase at bus 6.

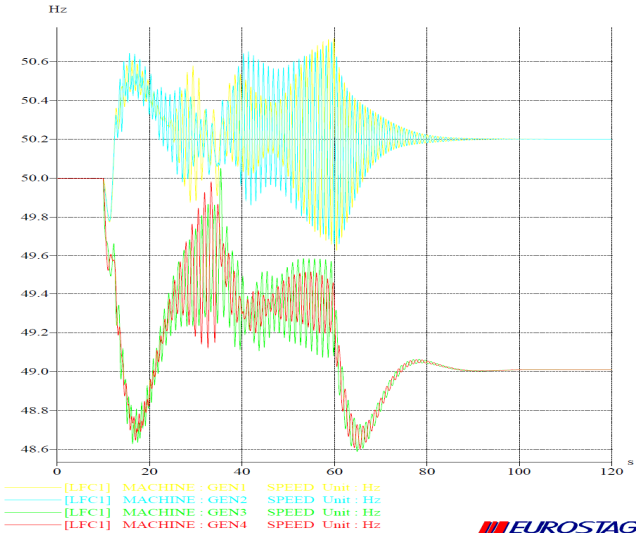


Fig. 16. Frequency after load shedding.

B. *PIAT Network*

In this part, the study of the frequency stability as well as load frequency control (LFC) was extended to the isolated Algerian network PIAT. In what follows, we present the simulation results following a supply-load unbalance. Figs.17 to 21 show the variation of system frequency for the following scenarios:

1. Load increase of 50%;
2. Load decrease of 50%;
3. Loss of load;
4. Loss of generator;
5. Loss of load and control application.

The simulation results with a load increase of 50% in Adrar area are shown in Fig.17.

The simulation results with a load decrease of 50% in In Salah (ISCH 1) area are shown in Fig.18.

The simulation results in case of a load loss in Timimoun area are shown in Fig.19.

The simulation results in case of a generator loss in Timimoun at bus 13 & 14 (30 MW) are shown in Fig.20.

The simulation results in case of a load loss in Timimoun area with AGC regulation applied to generators 17, 19 and 20 are shown in Fig.21.

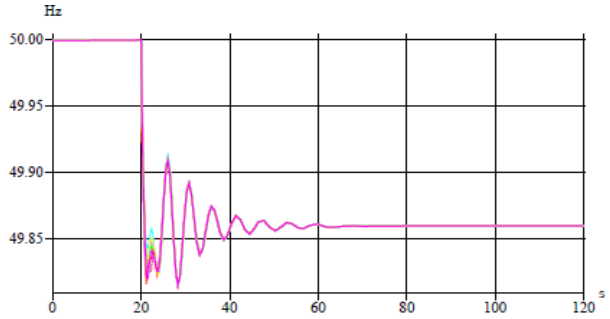


Fig. 17. Frequency following 50% load increase in Adrar.

According to Fig.17, there is two phases, a phase before the fault between t = [0 20]s, where the frequency of the network is stable f = 50Hz, and a second phase after the fault t =] 20 120]s, where we can observe a frequency drop that is due to the mechanical torque that is less than the resistive torque (Tm <Tr) which causes a decrease in the speed of rotation of the turbines.

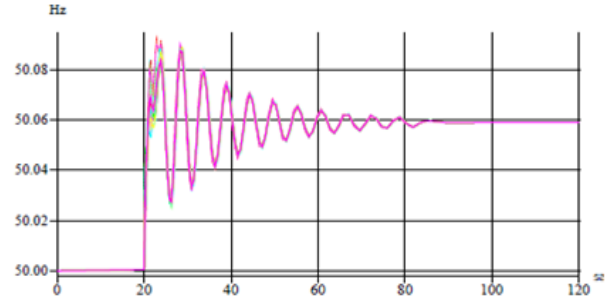


Fig. 18. Frequency following 50% load decrease in In Salah.

In this case, the mechanical torque is greater than the resisting torque (Tm> Tr), which leads to an increase in the speed of rotation of the turbines.

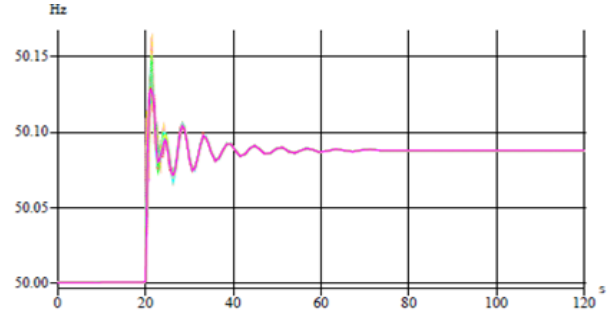


Fig. 19. Frequency following load loss in Timimoun.

In this case the generation is higher compared to consumption. The mechanical torque is greater than the resisting torque (Tm> Tr) so we will have an increase in the speed of rotation of the turbines.

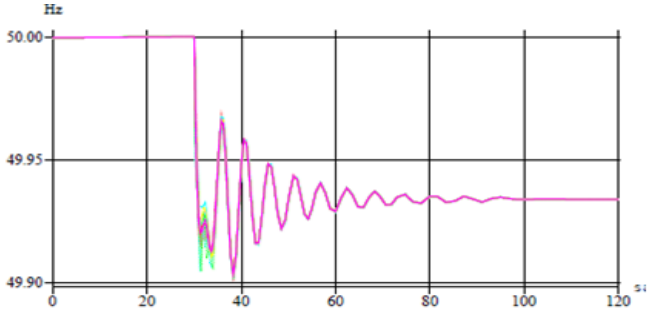


Fig. 20. Frequency following generator loss in Timimoun.

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In this case the production is lower than the consumption. The resistive torque is greater than the mechanical torque (*Tr*> *Tm*), which leads to a decrease in the speed of rotation of the turbines.

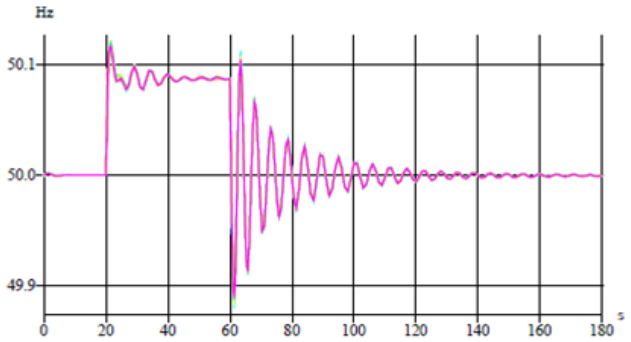


Fig. 21. Frequency following load loss in Timimoun and AGC.

Fig.21 shows the frequency variations of the 25 generators of the PIAT network following load loss and considering the AGC regulation applied on the selected units (17, 19 and 20). From this figure it can be observed the appearance of three phases, a phase before the fault, where the network frequency is stable at 50 Hz, and a second phase with a primary regulation, where it can seen an increase in frequency at 50.14 Hz because the mechanical torque is greater than the resisting torque (*Tm*> *Tr*) which lead to the increase in the speed of rotation of the turbines, thereafter a third phase, regulation phase, where the frequency stabilizes at its nominal value 50 Hz as a result of the action of secondary regulation of production.

V. Conclusion

In this work, power system stability and control issue using EUROSTAG software was investigated. A non-exhaustive presentation of the EUROSTAG software, with some models using the Macro-Block tool was presented. A dynamic time domain simulation of transient stability and frequency stability applied on the IEEE 10 bus system and the isolated PIAT Algerian network was performed. Moreover, some ancillary control action such as the AGC and UFLS have been also implemented to enhance the system stability. As future work, this study will be extended in presence of renewable energies such as wind farm and photovoltaic array with more scenarios and incorporating all AGC and UFLS stages.

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